

25

C

By conservation of moments,

magnitude of moment due to force acting on current carrying conductor =
magnitude of moment due to paper rider

$$\rightarrow B_{\text{coil}} IL(15) = W_{\text{rider}}(2)$$

$$\rightarrow \left(\frac{\mu_0 N I_{\text{coil}}}{2r} \right) IL(15) = W_{\text{rider}}(2)$$

6

$$\rightarrow \left(\frac{(4\pi \times 10^{-7}) 1000 I_{\text{coil}}}{2(12.5/100)} \right) (5)(4/100)(15/100) = \left(\frac{5}{1000} 9.81 \right) (2/100)$$

$$\rightarrow I_{\text{coil}} = 6.5 \text{ A}$$